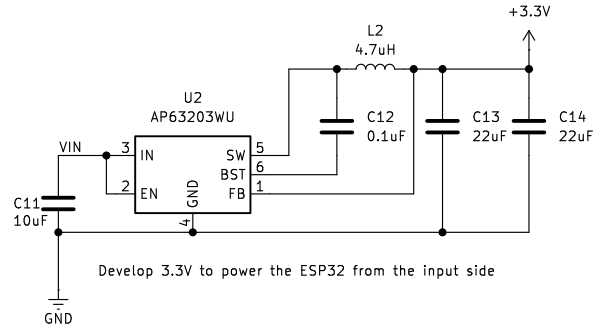
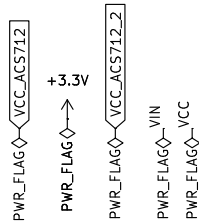


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Develop 3.3V to power the ESP32 from the input side

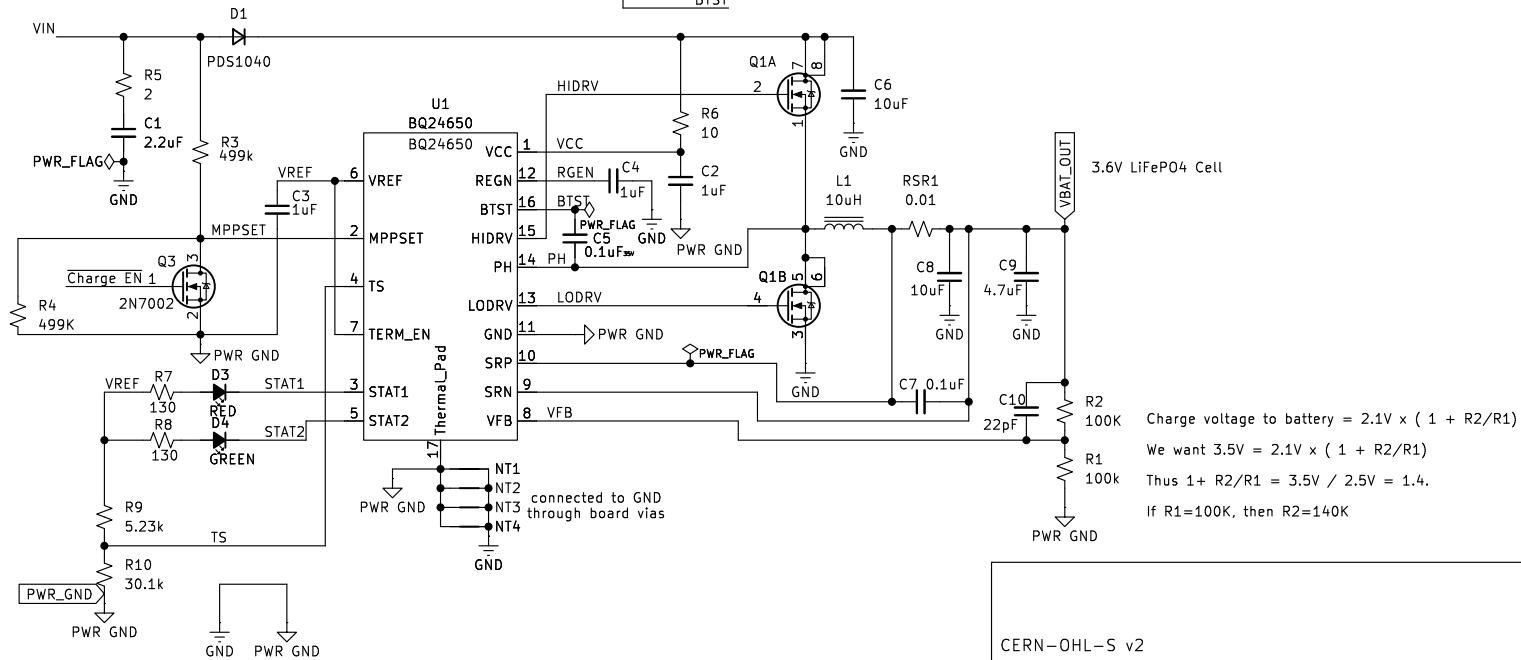
RSR1 sets max charge current:
 $I_{MAX} = 40\text{mOhms} / \text{RSR1}$

20mOhms thus limits to 2A

ACS712 on VCC=5V will have output on VIOUT to 3.3V ADC
 below 3.3V upto about 4A, or 7A if the 20A version of the ACS712 is used.

Thus we can use 10mOhms to allow upto 4A charging.
 With a 10Ah LiFePO4 is his 0.4C, which is below the
 save limit of 1.0C

ACS712 Actually requires VCC = 5V, not 3.3V
 according to datasheet?



Charge voltage to battery = $2.1V \times (1 + R2/R1)$

We want $3.5V = 2.1V \times (1 + R2/R1)$

Thus $1 + R2/R1 = 3.5V / 2.5V = 1.4$.

If $R1=100K$, then $R2=140K$

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Sheet: /

File: mppt-charge.kicad_sch

Title: Mega65Modular MPPT

Size: A4 Date: 2025-11-10

KiCad E.D.A. 9.0.2

Rev: v0.1

Id: 1/1